

SHIMANOVSK, Russian Federation

WMO: 314420

Lat: 51.9868N

Lon: 127.6530E

Elev: 281

StdP: 98.00

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.9	-35.6	-40.6	0.1	-37.6	-38.3	0.1	-35.1	6.9	-20.5	6.2	-19.9	0.1	340	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.5	19.7	27.8	19.3	26.2	18.7	22.1	26.3	21.0	25.1	20.1	24.2	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	15.9	24.1	19.7	14.9	22.9	18.7	14.0	22.0	66.5	26.2	62.7	25.0	59.4	24.3	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.5	5.7	-41.5	33.1	3.0	2.0	-43.6	34.5	-45.4	35.7	-47.0	36.8	-49.2	38.3		
(5)			-41.1	23.4	3.1	1.2	-43.3	24.2	-45.2	24.9	-46.9	25.6	-49.2	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-23.7	-19.6	-9.6	2.2	10.5	17.2	20.3	17.6	10.5	0.2	-13.7	-23.1
	DBStd	16.50	4.99	5.06	6.17	4.64	4.15	3.49	2.73	3.15	4.49	5.30	6.66	5.14
	HDD10.0	4853	1044	828	608	237	43	1	0	1	48	306	711	1026
	HDD18.3	7125	1303	1062	866	483	246	60	11	52	236	562	961	1284
	CDD10.0	901	0	0	0	4	58	217	321	235	63	2	0	0
	CDD18.3	131	0	0	0	0	3	26	73	28	1	0	0	0
	CDH23.3	1249	0	0	0	4	61	344	588	234	18	1	0	0
	CDH26.7	299	0	0	0	1	13	97	150	38	1	0	0	0
Wind	WSAvg	2.3	1.7	1.9	2.6	3.1	2.9	2.2	2.0	2.1	2.4	2.6	2.1	1.9
Precipitation	PrecAvg	512	6	5	9	23	57	72	127	94	59	26	13	9
	PrecMax	841	21	20	54	89	120	136	292	225	111	91	37	18
	PrecMin	389	1	0	0	0	10	16	52	19	18	3	0	0
	PrecStd	99	4	4	11	20	27	35	66	41	27	19	9	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.9	-3.0	8.8	22.7	28.5	32.6	32.9	29.5	25.3	18.3	4.5	-5.4
		MCWB	-10.2	-5.9	2.9	10.9	15.6	20.0	20.8	20.7	16.6	10.6	0.5	-7.2
	2%	DB	-12.0	-5.5	5.4	17.0	24.7	29.2	29.9	27.7	23.2	13.9	0.9	-9.9
		MCWB	-13.2	-7.4	0.5	7.9	13.6	18.6	20.6	19.9	16.0	7.8	-1.4	-11.0
	5%	DB	-14.0	-8.4	2.8	13.5	22.0	27.0	28.1	25.9	20.7	10.9	-1.3	-13.1
		MCWB	-15.0	-10.1	-1.2	6.3	12.1	17.7	20.2	19.0	14.5	6.0	-3.2	-14.1
	10%	DB	-16.0	-11.0	0.2	10.7	19.4	25.0	26.5	24.3	18.4	8.3	-3.9	-15.5
		MCWB	-16.8	-12.4	-2.6	4.5	11.0	16.9	19.8	18.4	13.4	4.2	-5.4	-16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.1	-5.5	3.1	11.3	16.3	22.7	23.4	22.7	18.6	11.6	1.1	-7.1
		MCDB	-8.8	-3.5	8.6	21.0	25.5	28.4	27.9	26.9	22.1	17.2	3.9	-5.3
	2%	WB	-13.0	-7.4	0.7	8.5	14.7	20.1	22.5	21.4	17.1	8.6	-1.1	-11.0
		MCDB	-12.0	-5.4	4.4	16.0	22.5	26.1	26.8	25.0	21.2	12.7	0.9	-10.1
	5%	WB	-14.9	-10.0	-0.9	6.6	13.2	18.9	21.6	20.4	15.7	6.4	-3.2	-14.0
		MCDB	-14.0	-8.5	2.8	12.9	19.8	24.7	25.8	24.0	19.2	10.4	-1.3	-13.1
	10%	WB	-16.8	-12.3	-2.7	4.8	11.9	17.9	20.7	19.4	14.4	4.5	-5.3	-16.2
		MCDB	-16.0	-11.0	0.2	10.2	17.8	23.2	24.8	22.8	17.8	7.9	-3.9	-15.5
Mean Daily Temperature Range	5% DB	MDBR	11.8	13.9	12.8	11.7	12.6	12.3	10.5	11.0	12.1	10.9	10.9	10.9
		MCDBR	11.6	15.9	14.3	17.3	18.3	16.4	14.0	14.4	15.6	15.4	11.9	10.9
		MCWBR	10.6	13.9	10.1	10.1	8.9	7.1	6.0	7.0	8.2	9.7	9.3	10.0
	5% WB	MCDBR	11.6	15.5	13.3	16.0	15.0	13.3	10.6	10.8	12.3	13.7	11.5	10.8
		MCWBR	10.6	13.6	9.4	9.6	7.5	6.7	5.3	5.9	7.1	9.2	9.1	9.9
Clear-Sky Solar Irradiance	taub	0.259	0.293	0.370	0.551	0.621	0.502	0.474	0.417	0.389	0.380	0.308	0.264	
	taud	2.160	2.088	1.927	1.666	1.610	1.991	2.126	2.278	2.268	2.121	2.135	2.110	
	Ebn at Noon	737	808	796	682	659	762	778	808	784	687	649	647	
	Edh at Noon	70	102	149	221	245	169	145	118	106	100	73	62	
All-Sky Solar Radiation	RadAvg	1.46	2.65	4.34	5.14	5.38	6.04	5.42	4.75	3.71	2.43	1.52	1.12	
	RadStd	0.05	0.07	0.20	0.53	0.41	0.65	0.48	0.32	0.29	0.15	0.07	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page